Project 5: Naive Bayes Classifier

By Spencer Hardy and Zach Maughan

Git: Git Repo
Slides: Slides

Introduction

The card game Magic: The Gathering has five colors, with each color having its own set of abilities. For instance, the color “blue” is typically the best at drawing more cards, while “green” is the best at gathering resources. This concept is known as the “color pie” and it drives the core design principles for the game. There have been over 30,000 unique cards printed over three decades, and although occasionally a card is designed which breaks the norm of that color, we still wanted to see if the rules have been followed consistently enough over the game's lifespan that we could predict a card’s color just from what the card does.

Using card data from Scryfall, a large community card database, and Naive Bayes classifiers to predict a card’s color two ways, the “oracle tags” which describe what a card does, and from the card’s rules text. Our results showed that a card’s color can be predicted with decent performance, but this process can also reverse engineer the “color pie” and show which types of abilities belong to each color. This may be useful to card designers because it can potentially expose examples of “good” and “bad” card design in the context of the color pie.

Dataset

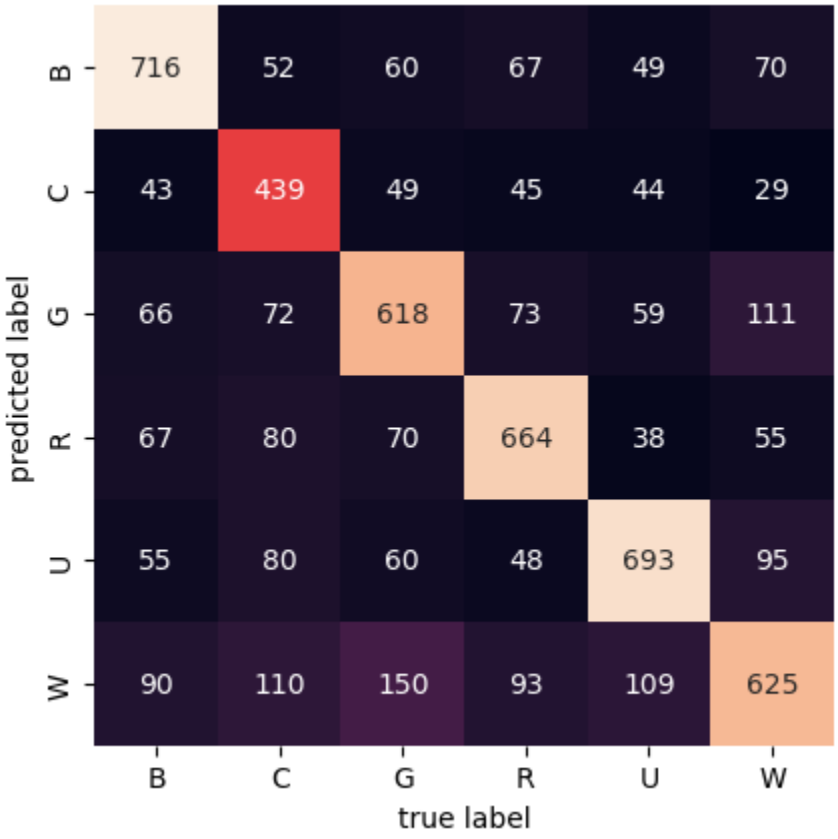
We used two bulk files downloaded from Scryfall, a card database website run by the community. The first file is “oracle cards”, with each row having a distinct card with name, color, and official rules text. The second is “oracle tags”, which is built from a community effort to label cards by the functions of the cards. There are thousands of tags ranging in description with each card having roughly eight tags on average. Using both of these makes for a good fit for the question because the rules text is the hard truth of what the card does, and the tags are roughly the same thing but with a lot of noise removed. We removed lands, a type of card which generates resources, because a vast majority of them aren’t complex enough to adhere to the color pie. We also removed cards without any rules text, and multicolored cards. This left us with all five colors in the game, plus colorless, totaling six categories, which are roughly equally distributed among the population of 30,000 cards.

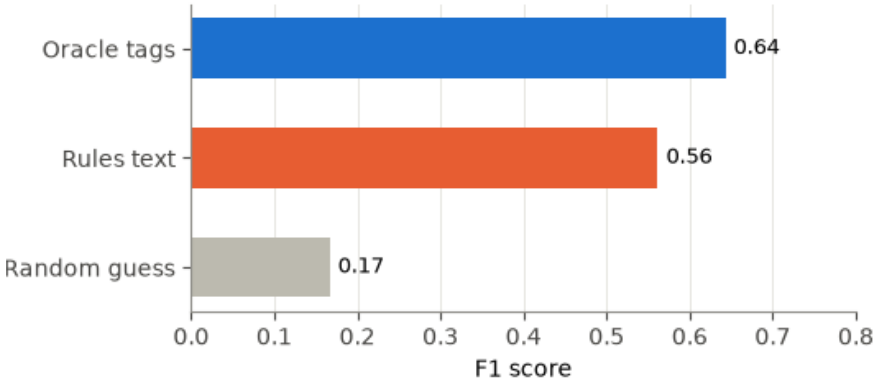
Analysis Technique

We used a multinomial Naive-Bayes classifier. A classification model is necessary because we are trying to predict color, a categorical variable. Naive-Bayes is often used for text analysis, such as spam detection or sentiment analysis, which is perfect for what we are trying to do. The oracle text for the cards is in the format we want it to be already, meanwhile the oracle tags can be interpreted in a very similar way to word appearances.

Results

We found that our classification model trained on oracle tags can predict a card's color with a precision of 0.6474, recall of 0.6425, and F1 score of 0.6428. The confusion matrix for this model can be seen below. Meanwhile, the same type of model trained on oracle text achieved a precision score of 0.5682, recall of 0.5619, and F1 score of 0.5615. We compared this to a random classifier. Because there are 6 categories (5 colors + colorless), and the cards are roughly equally distributed between all 6, we simply gave the random classifier an F1 score of ⅙ as an approximation.





We also looked at many model's predictions with which it had the greatest confidence, both correct and incorrect predictions. These predictions can be useful case studies for card

designers, as they represent extreme examples of cards that perfectly fit into the color pie, and ones that ignore it completely. Furthermore, we determined the tags and words most commonly associated with each color, effectively providing a reconstruction of the rules of the color pie, and giving card designers more insight into how to best design their cards.

Technical

Both files we pulled from Scryfall were in a JSONL format and loaded them with pandas. From the cards we dropped anything with “Land” in its type line and kept oracle_id, name, colors and oracle_text, mapped empty color lists to “C”, single colors to their letter, and dropped multicolor cards. Some cards are double faced (printed on front and back), these cards had no top-level color so these were also dropped. For the tags, we inverted Scryfall’s tag-to-cards list into a cards-to-tag dictionary and made a sparse binary card-by-tag matrix, this reduces overhead since a dense version with thousands of tags would be huge and mostly zeros. For the text model, we replaced newlines, filled missing text with empty strings and used CountVectorizer. Both models had a train-test split of 80:20, and both used the same parameters for the Naive-Bayes classifier.

Limitations / Future Direction

If we had more time to work on this project, we would try to find the most common words and the most common tags that appear among all colors, which only serve as filler, and remove them from the data. In text analysis it’s very common to get a list of “stop words” and do this, but in this case if we did something similar, it would likely reduce noise and improve results.

Another limitation to this study is the fact that the oracle tags are sourced from a community effort to go through and manually label each card with the relevant tag. However, the creation and assignment of these tags is fairly subjective, which may influence results.